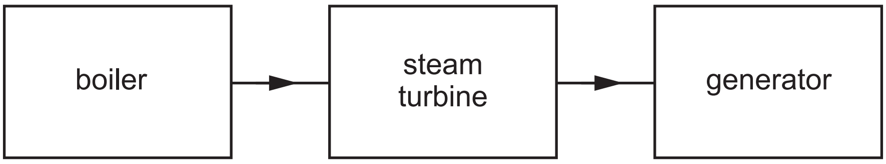
Examiner only

2. The block diagram below represents a fossil fuel power station.



(a) (i) Use the words in the block diagram to explain how electricity is generated in a fossil fuel power station. [3]

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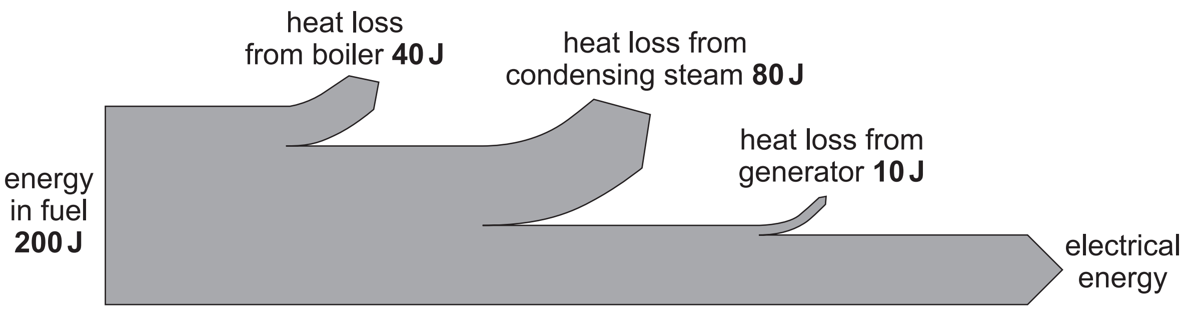
(ii) Neve says that fossil fuel power stations are about 35% efficient.
Jacob says they are less than 30% efficient.
Use the information in the Sankey diagram below and an equation from page 2 to explain whether you agree with Neve or Jacob. [3]



Space for calculation.

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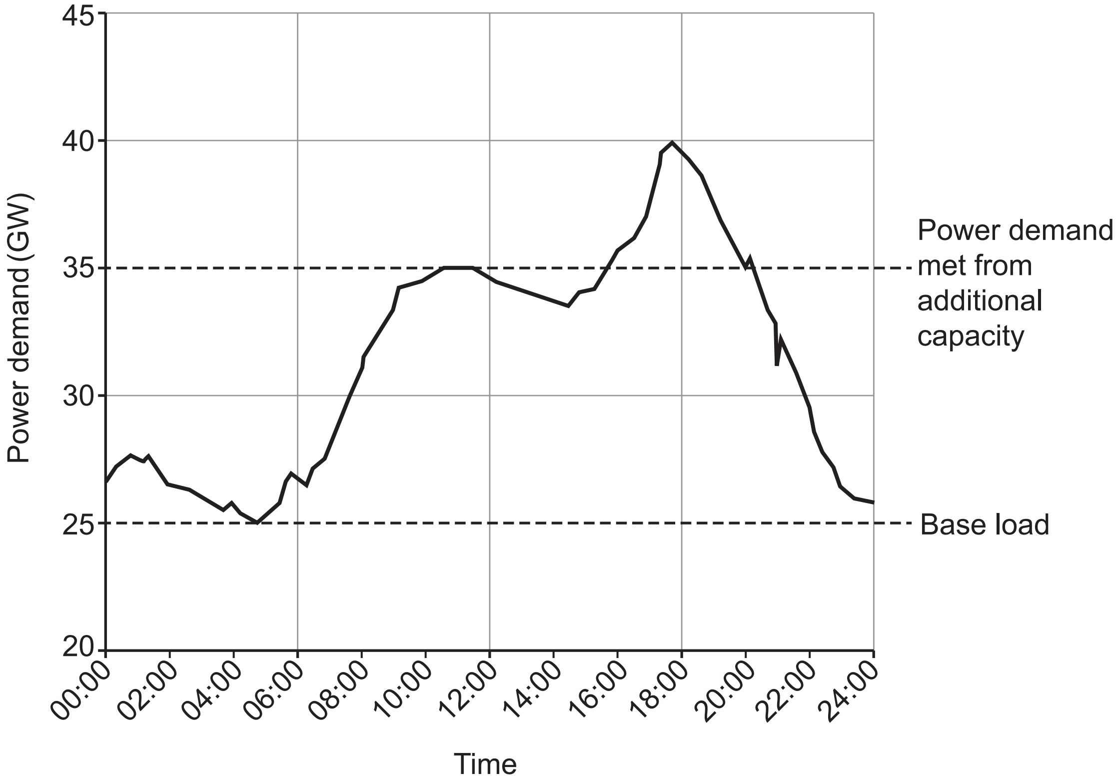
(b) Power stations are connected to the National Grid.
The power demand on a particular day is shown below.



(i) The graph shows that the base load is 25 GW. State what is meant by this. [1]

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(ii) State the peak power demand on the National Grid on the day shown in the graph. [1]

peak power demand = ... GW

Examiner only

(iii) The Dinorwig power station is a pumped storage hydro-electric scheme.

Dinorwig has six 0.3 GW generators.
It is used to provide additional power once demand rises above 35 GW.
Once operating at maximum output, it can provide power for 6 hours before running out of water.

Rowan says that Dinorwig can provide power for the whole time that demand is over 35 GW.
However, he says that it will not be able to supply enough power to meet all the demand above 35 GW shown in the graph.
He also says that about 3 GW will need to be imported from overseas.

Explain whether you agree with Rowan.
Include data in your answer. [4]

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